Discussion Questions

1) A pro of event driven programming is that the program only responds when something happens, making it more efficient, however the con is that it is harder to follow and debug if there is an issue.
2) The difference between the two mechanisms is that the TTL limits how long a packet can travel and the flooding checks stop nodes from forwarding the same packet repeatedly, using both provides two layers of protection. If you only use TTL duplicate packets would be sent repeatedly until the TTL hits zero thus would use more bandwidth. If you only use the flooding checks it can lead to memory exhaustion and loops.
3) The best case is that the destination is a direct neighbor so only one packet is sent. The worst case is a topology, such as a full mesh topology, where every node receives and floods the packet again.
4) A better method is building a routing table using neighbor information and only sending packets to the next hop.
5) A different design could be to store only the highest sequence number rather than a cache of source and sequence number thus using less memory.